

SemVideo: Reconstructs What You Watch from Brain Activity via Hierarchical Semantic Guidance

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Abstract

Reconstructing dynamic visual experiences from brain activity provides an avenue for exploring the neural mechanisms of visual perception. While progress in fMRI-based image reconstruction has been notable, extending this success to video reconstruction remains a challenge. Current fMRI-to-video reconstruction approaches encounter two major shortcomings: (i) inconsistent visual representations of salient objects across frames, leading to appearance mismatches; (ii) poor temporal coherence, resulting in motion misalignment or abrupt frame transitions. To address these limitations, we introduce **SemVideo**, a fMRI-to-video reconstruction framework guided by hierarchical semantic information. At the core of SemVideo is **SemMiner**, a hierarchical guidance module that constructs three levels of semantic cues from the video stimulus: static anchor descriptions, motion-oriented narratives, and holistic summaries. Leveraging this semantic guidance, SemVideo comprises three key components: a Semantic Alignment Decoder that aligns fMRI signals with CLIP-style embeddings derived from SemMiner, a Motion Adaptation Decoder that reconstructs dynamic motion patterns using a novel tripartite attention fusion architecture, and a Conditional Video Render that leverages hierarchical semantic guidance for video reconstruction. Experiments on the CC2017 and HCP datasets demonstrate that SemVideo achieves superior performance in semantic alignment and temporal consistency, setting a new state-of-the-art in fMRI-to-video reconstruction. The code is publicly available at <https://github.com/yang-minghan/SemVideo>.

1. Introduction

Understanding the information encoded in brain activity remains one of the central challenges in cognitive neuro-

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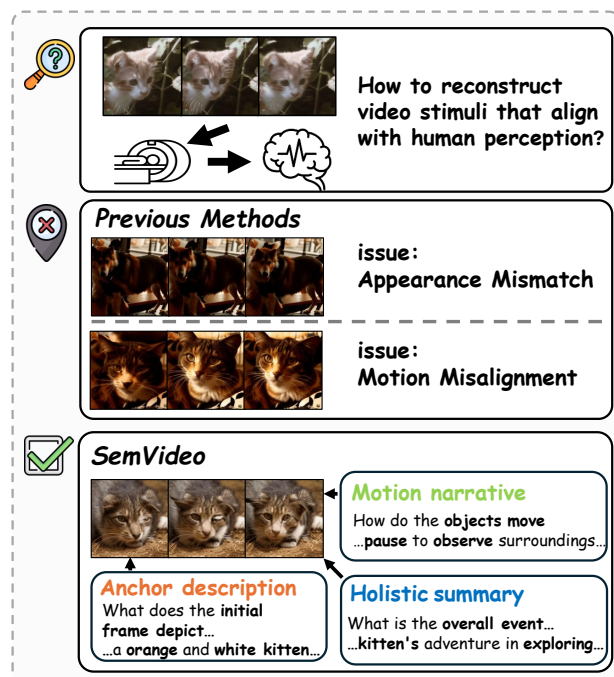


Figure 1. **Top:** While a subject watches an original video stimulus, their brain activity is recorded via fMRI. **Middle:** Reconstructed results from previous methods, which suffer from two issues: *Appearance Mismatch* and *Motion Misalignment*. **Bottom:** Reconstructed result from SemVideo, achieves both semantic consistency (reconstructing the “kitten”) and motion coherence (matching dynamic actions like “crouching” and “turning”) by leveraging hierarchical semantic descriptions as intermediate targets to guide the fMRI signal decoding process.

science. Reconstructing external stimuli from brain signals — particularly using non-invasive methods such as functional Magnetic Resonance Imaging (fMRI) — is an important yet highly challenging task. Neural decoding typically encompasses three fundamental tasks[19]: classification, identification, and reconstruction. With the rapid

advancement of deep learning techniques, prior work has made remarkable progress in both the classification[29, 77] and identification[26, 30] of static visual stimuli from fMRI data. Notably, the release of the Natural Scenes Dataset (NSD)[1]—the largest publicly available fMRI-image paired dataset — has catalyzed significant breakthroughs in image reconstruction[8, 16, 58]. Moreover, the emergence of powerful text-to-image diffusion models has brought high-quality, high-resolution image reconstruction from fMRI signals within reach[9, 28, 40, 45, 50, 69, 70, 73, 74], transforming what was once considered an unattainable goal into a practical possibility.

However, reconstructing *dynamic visual stimuli* from fMRI signals remains a significant challenge due to the inherent limitations of fMRI, which relies on the slow hemodynamic response of BOLD signals. These signals integrate brain activity over several seconds, making it difficult to capture the rapid motion variations present in video stimuli. Consequently, fMRI-based video reconstruction becomes an even more complex and challenging task. Despite this, the development of video generation technology has greatly enhanced the feasibility of fMRI-based video reconstruction tasks[10, 13, 15, 41, 60]. However, two major issues remain unresolved in this task: (i) inconsistent visual representations of salient objects across frames, leading to appearance mismatches, and (ii) poor temporal coherence, resulting in motion misalignment or abrupt frame transitions.

In this work, inspired by neuroscience research indicating that the human brain perceives videos discretely[51, 56] due to phenomena such as persistence of vision[6, 7, 12, 46, 82] and delayed memory[32, 38, 65]. This makes it impractical for brain signals to process every video frame and every pixel; instead, only keyframes elicit significant responses from the brain’s visual system. This suggests that focusing on key semantic perceptions, rather than exhaustive pixel-by-pixel analysis, aligns more closely with the efficiency and practicality of the human visual system.

To this end, we first propose *SemMiner*, a module built upon a multimodal large language model (MLLM [47–49, 63]), which decomposes the original video stimuli into fine-grained, multi-level textual descriptions including holistic summaries, static anchor descriptions, and motion-oriented narratives. This hierarchical supervision better simulates the way humans process and recall visual experiences, addressing the semantic under-specification problem in traditional pipelines. Guided by this hierarchical supervision, we then present *SemVideo*, a novel fMRI-to-video reconstruction framework guided by hierarchical semantic supervision and motion-aware latent decoding. Our framework is built upon three core components: (i) **Semantic Alignment Decoder**. To decode hierarchical semantic features from fMRI signals and adapt to the variable signal dimensions across different subjects, we pro-

pose the Semantic Alignment Decoder (SAD). This cross-subject multi-level semantic decoder consists of a subject-specific projector, a subject-shared mapper, and a Refiner module, enabling precise semantic feature decoding from fMRI signals while minimizing noise. (ii) **Motion Adaptation Decoder**. To reconstruct coherent action sequences from brain signals, we designed the Motion Adaptation Decoder (MAD). The core of the MAD is a tripartite attention fusion architecture that synergistically integrates spatial self-attention, temporal self-attention, and semantic-guided cross-attention mechanisms. By explicitly injecting semantic priors into the attention computation, this module effectively aligns motion latents with both spatial structures and semantic actions, enhancing the fidelity and continuity of the reconstructed video dynamics. (iii) **Multi-Stage Semantic Fusion for Video Reconstruction**. We propose a multi-level, staged guidance strategy for conditional video generation. Instead of relying on a single semantic cue, we progressively condition the generation on three sources: static semantics (anchor descriptions), dynamic semantics (motion-oriented narratives), and holistic video-level semantics (holistic summaries). This structured conditioning ensures that the synthesized video aligns with both the static and temporal structure of the original perceptual input, yielding more coherent and realistic outputs.

Extensive experiments have demonstrated the superior performance and robustness of *SemVideo* on two widely-used public datasets, CC2017 [71] and HCP [42]. We evaluate our method against previous approaches across three dimensions: semantic-level, pixel-level, and spatiotemporal-level metrics, achieving SOTA performance on 8 out of 10 metrics. Additionally, we provide a neuroscience-based interpretation of the effectiveness of *SemVideo* using ROI-wise visualization techniques, which confirm that our hierarchical semantic descriptions activate corresponding brain regions and validate the model’s effectiveness.

2. Related Work

2.1. Video Caption Generation

To introduce richer semantic guidance, a recent research [9, 18, 57, 58, 62, 69, 76] trend has shifted from unimodal visual feature alignment to dual-modal vision-text alignment. However, the effectiveness of this strategy is constrained by a significant bottleneck: the scarcity of high-quality, open-source video captioning models. Lacking dedicated video captioning tools, current research [13, 15, 41] often resort to applying Image Captioning models, such as GIT [68], BLIP [36, 37, 75] and SmallCap [53]), to discrete video frames. This method generates a series of short, static descriptions that fail to capture both the video’s temporal dynamics and its fine-grained semantic details. Consequently, even when supervised by this textual modality, downstream

reconstruction models are unable to acquire comprehensive information about either motion or semantics. This dual deficiency is a primary cause of persistent issues with motion coherence and semantic accuracy in generated videos, representing a core challenge for the field.

2.2. Diffusion-based video generation

The feasibility of leveraging the fine-grained semantic guidance for the reconstruction task is critically dependent on the recent advancements in video generation. Building on the remarkable progress in image synthesis with diffusion models [22, 52, 55], the field has rapidly advanced to text-to-video (T2V) diffusion models [4, 25, 59]. These approaches, including 3D diffusion models [24] and methods such as AnimateDiff [20], excel at synthesizing temporally coherent motion strongly conditioned by textual descriptions. This ability to translate semantic intent into dynamic visual content positions them as the ideal generative backbone for fMRI-to-video decoding tasks.

2.3. fMRI-based Video Reconstruction

Early approaches [21, 34, 35, 44, 66, 71] to video reconstruction treated the task as a sequence of independent image reconstructions. These methods employed Generative Adversarial Networks (GANs [17]) or Autoencoders (AEs [33]) to synthesize frames from brain signals, with the primary goal of recovering low-level visual features. However, reconstructions from these models often lacked discernible semantic content. In contrast, recent studies have largely shifted toward generative frameworks centered on diffusion models [20, 23, 24]. One line of work, including Brain Netflix [13] and MinD-Video [10], utilizes masked-brain modeling (MBM) to map fMRI signals into a unified latent space that drives a video diffusion model. Another class of methods [15, 41, 60] focus on aligning fMRI signals with the deep features from pretrained models—such as CLIP embeddings or VAE encodings of video frames—to guide video reconstruction. Despite significant improvements in quality, these advanced methods still exhibit two primary limitations: insufficient motion coherence and semantic deviation. This gap highlights a pressing need for a more fine-grained semantic supervision mechanism to guide the video generation process.

3. Methodology

In this section, we present *SemMiner* and *SemVideo*, two key components of our proposed fMRI-to-video reconstruction method. *SemMiner* deconstructs a given video into fine-grained, multi-level textual descriptions, organizing them into three hierarchical semantic perspectives: first-frame appearance, motion cues, and overall temporal consistency. *SemVideo* is a decoding framework that transforms fMRI signals into semantic embeddings and motion latents,

which are then used as conditional inputs for a text-to-video generation model to reconstruct the original video stimuli.

3.1. SemMiner

To provide precise and diverse semantic supervision for fMRI-to-video reconstruction, we introduce *SemMiner*, a hierarchical guidance module constructs three levels of semantic cues from the original video stimulus. Given a video stimulus V , these mined descriptions are structured into three complementary semantic perspectives. The **anchor description** C_{anchor} captures the static visual content of the first frame, serving as a semantic anchor to ensure basic alignment between the reconstructed and original videos. The **motion-oriented narratives** C_{motion} focus on fine-grained dynamic cues, particularly highlighting actions and dynamic transitions within the video. Finally, the **holistic summaries** C_{holi} encapsulate a global semantic summary of the entire video, integrating both static and motion information into a coherent narrative. These descriptions establish a comprehensive framework, providing multi-level semantic guidance for the reconstruction.

SemMiner is implemented as a two-stage semantic decomposition module. In the first stage, it generates a simple semantic summary C_{basic} of V using a MLLM Ψ [2, 79, 80], constrained to a maximum of 20 words, formulated as:

$$C_{\text{basic}} = \Psi(P_{\text{basic}}, V), \quad (1)$$

where P_{basic} denotes the initial instruction used for synthesizing the basic description. This short summary serves as a “rein”, to prevent the subsequent generation of diverse captions from becoming overly unconstrained or drifting semantically—analogueous to controlling a “runaway horse” [27, 31, 39, 81]. The imposed length limit ensures that C_{basic} captures only the most essential content, thereby reducing the risk of introducing strong priors that could dominate later stages.

In the second stage, *SemMiner* is steered by carefully crafted instructions P_L corresponding to specific semantic targets. Conditioned on the video V and the initial summary C_{basic} , the framework elicits the target-specific descriptions:

$$C_L = \Psi(P_L, C_{\text{basic}}, V), \quad L \in \{\text{anchor, motion, holi}\}, \quad (2)$$

where P_L denotes the instruction specifically designed for each semantic objective. The detailed instructions used in *SemMiner* can be found in Appendix A.1.

CC2017-SE. To advance the fMRI-to-video reconstruction field, we introduce CC2017-SE, a semantic extension of the CC2017 dataset. Building upon the original dataset, we apply *SemMiner* to generate three semantic descriptions — C_{anchor} , C_{motion} , and C_{holi} . We also provide quantitative and qualitative analysis of synthesized captions in Appendix A.2. The extended dataset is available at <https://huggingface.co/datasets/YanmHa/SemVideo>.

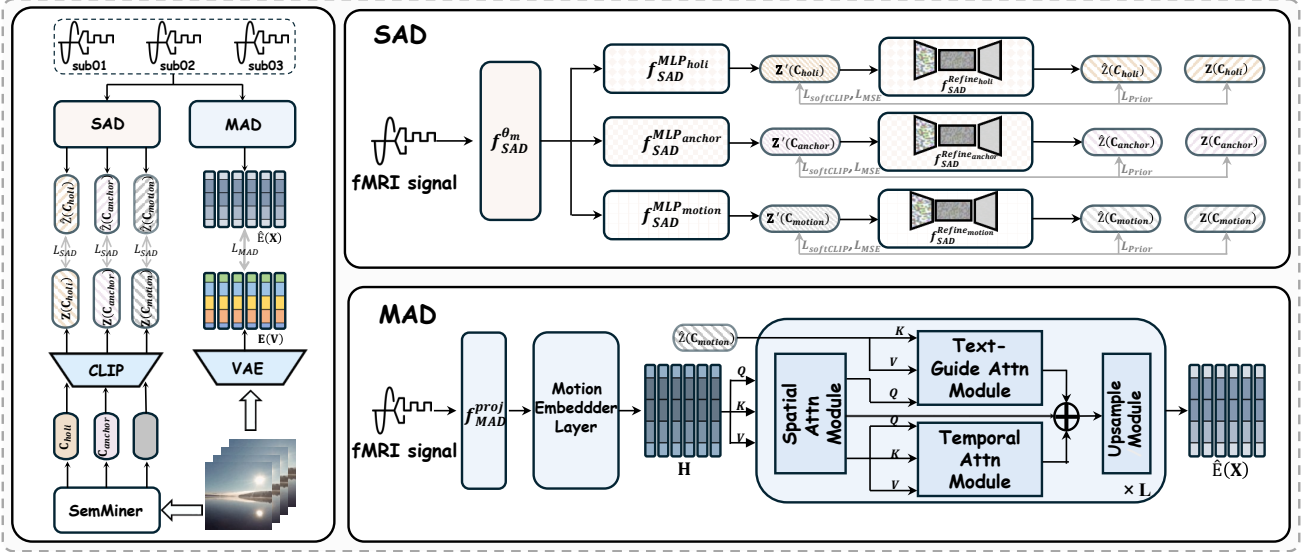


Figure 2. Overview of the *SemVideo* training pipeline. In the first stage, the Semantic Alignment Decoder is trained to map fMRI signals to three levels of semantic targets, denoted as $Z(C_L)$. In the second stage, the Motion Adaptation Decoder is trained to utilize the predicted motion semantics $\hat{Z}(C_{\text{motion}})$ to refine the latent embedding of each reconstructed frame.

3.2. SemVideo

SemVideo operates in three key stages, as illustrated in Figure 2 and Figure 3. First, the fMRI signals $X \in \mathbb{R}^{D_m}$ are decoded into semantic feature representations $\hat{Z}(C_L) \in \mathbb{R}^{77 \times 768}$, where D_m denotes the number of activated voxels for subject m , and $Z(\cdot)$ represents the target feature space. Second, it reconstructs a motion-guided latent representation of the video, denoted as $\hat{E}(X)$, guided by the decoded motion semantics $\hat{Z}(C_{\text{motion}})$. Finally, the decoded semantic and motion features are fused to generate the reconstructed video via Conditional Video Render Φ , formulated as:

$$\hat{V} = \Phi(\hat{Z}(C_{\text{anchor}}), \hat{Z}(C_{\text{holi}}), \hat{E}(X)). \quad (3)$$

Semantic Alignment Decoder (SAD) The objective of the Semantic Alignment Decoder (SAD) is to take the fMRI signal X as input and output the predicted semantic feature $\hat{Z}(C_L)$. Since the number of activated voxels varies across subjects, SAD first applies a subject-specific projection layer $f_{\text{SAD}}^{\theta_m}$ to project X into a unified latent space of dimension D_l , yielding $X' = f_{\text{SAD}}^{\theta_m}(X) \in \mathbb{R}^{D_l}$. Next, a subject-shared encoder f_{SAD} maps the normalized fMRI representation X' to the CLIP text feature space, resulting in $\hat{Z}(C_L) = f_{\text{SAD}}(X')$. The encoder f_{SAD} is composed of a four-layer multilayer perceptron ($f_{\text{SAD}}^{\text{MLP}}$) followed by a Refineformer ($f_{\text{SAD}}^{\text{Refine}}$), a causal transformer designed to maximize the extraction of meaningful neural activity while minimizing the influence of noise, thereby enhancing the alignment between fMRI signals and the target semantic space. The training of SAD is supervised by a combination of three

training objectives:

$$\mathcal{L}_{\text{MSE}} = \|Z(C_L) - f_{\text{SAD}}^{\text{MLP}}(X')\|^2, \quad (4)$$

$$\mathcal{L}_{\text{SoftCLIP}} = - \sum_{j=1}^B \left[\frac{\exp\left(\frac{Z(C_L) \cdot Z(C_L^j)}{\tau}\right)}{\sum_{k=1}^B \exp\left(\frac{Z(C_L) \cdot Z(C_L^k)}{\tau}\right)} \times \log\left(\frac{\exp\left(\frac{f_{\text{SAD}}^{\text{MLP}}(X') \cdot Z(C_L^j)}{\tau}\right)}{\sum_{k=1}^B \exp\left(\frac{f_{\text{SAD}}^{\text{MLP}}(X') \cdot Z(C_L^k)}{\tau}\right)}\right) \right], \quad (5)$$

$$\mathcal{L}_{\text{refine}} = \mathbb{E}_{t \sim [1, T]} \left\| f_{\text{SAD}}^{\text{Refine}}\left(Z_L^t, t, f_{\text{SAD}}^{\text{MLP}}(X')\right) - Z(C_L) \right\|^2, \quad (6)$$

$$\mathcal{L}_{\text{SAD}} = \lambda_{\text{refine}} \mathcal{L}_{\text{refine}} + \lambda_{\text{SoftCLIP}} \mathcal{L}_{\text{SoftCLIP}} + \mathcal{L}_{\text{MSE}}, \quad (7)$$

where B denotes the batch size, j indexes the j -th sample in the batch, and τ is a temperature hyperparameter. T represents the total number of denoising timesteps, and Z_L^t is the perturbed semantic embedding of C_L at timestep t . λ_{refine} and $\lambda_{\text{SoftCLIP}}$ are weighting coefficients that balance the contributions of the corresponding loss components.

Motion Adaptation Decoder (MAD) To improve the fidelity of motion dynamics reconstructed from brain signals, we propose a semantically guided motion adaptation decoding strategy. Let the original video stimulus be denoted as $V = \{v_1, v_2, \dots, v_n\}$, where n is the number of frames. We extract its motion latent representation using a pretrained VAE encoder $\mathbf{E}$, yielding $E(V) = \{e_1, e_2, \dots, e_n\}$.

The objective of MAD is to take the fMRI signal X as input and generate a sequence of predicted motion la-

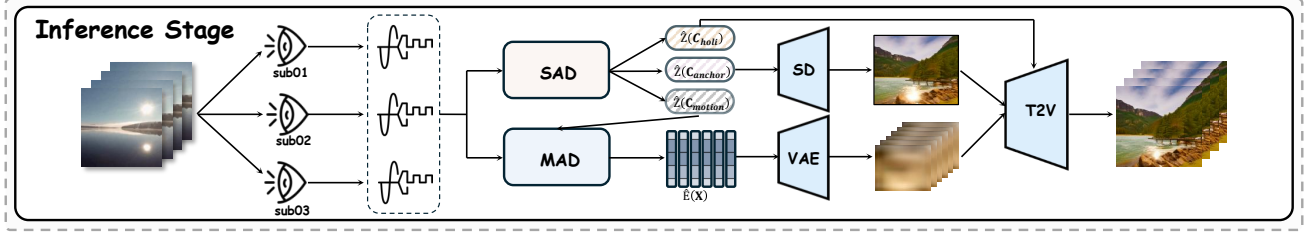


Figure 3. Overview of the *SemVideo* inference pipeline. fMRI signals are first decoded into $\hat{Z}(C_L)$ by the SAD. $\hat{Z}(C_{\text{motion}})$ conditions the MAD to refine frame embeddings $\hat{E}(x)$, which are passed through a VAE decoder, generating a blurry video. $\hat{E}(x)$ and $\hat{Z}(C_{\text{anchor}})$ guide the SD model to generate anchor frame, combined with the blurry video and $\hat{Z}(C_{\text{holi}})$, is fed into a T2V model, yielding final reconstruction.

tents corresponding to the reconstructed video frames, denoted as $\hat{E}(X) = \{\hat{e}_1, \hat{e}_2, \dots, \hat{e}_n\}$. To achieve this, MAD first projects the fMRI signal into the latent space using a subject-specific projection network $f_{\text{MAD}}^{\text{proj}}$, implemented as a multilayer perceptron¹. This projection yields a latent tensor $g = f_{\text{MAD}}^{\text{proj}}(X) \in \mathbb{R}^{n \times D_{\text{emb}}}$, where D_{emb} denotes the embedding dimension. Next, the tensor g is processed by a Motion Embedder Layer to produce a sequence of embeddings $S = \{s_1, s_2, \dots, s_n\} \in \mathbb{R}^{n \times c \times h \times w}$, where $c \times h \times w$ is the dimension of the embedding latent space. This embedding sequence serves as the initial input to the subsequent attention-based decoding module. MAD conducts a hierarchical fusion attention mechanism:

(i) To capture intra-frame structure, we apply *spatial attention* to each s_i . The latent tensor is reshaped and attended via scaled dot-product attention:

$$\hat{S} = \text{Reshape}(S) \in \mathbb{R}^{(n \cdot h \cdot w) \times c}, \quad (8)$$

$$E_{\text{spat}} = \text{Softmax} \left(\frac{Q_{\text{spat}} K_{\text{spat}}^\top}{\sqrt{d}} \right) V_{\text{spat}}, \quad (9)$$

where $Q_{\text{spat}} = \hat{S} W_{Q_{\text{spat}}}$, $K_{\text{spat}} = \hat{S} W_{K_{\text{spat}}}$, $V_{\text{spat}} = \hat{S} W_{V_{\text{spat}}}$.

(ii) To model inter-frame dependencies, we apply *temporal attention* across the temporal axis. We first flatten each spatially attended frame into a temporal sequence:

$$\tilde{S} = \text{Reshape}(E_{\text{spat}}) \in \mathbb{R}^{(c \cdot h \cdot w) \times n}, \quad (10)$$

The attention computation follows the same formulation as in the spatial attention and is omitted here for brevity.

(iii) We further incorporate *semantic guided cross-attention*. Let $\hat{Z}(C_{\text{motion}})$ be the predicted motion descriptions from the video, as formulated by:

$$E_{\text{cross}} = \text{Softmax} \left(\frac{Q_{\text{cross}} K_{\text{cross}}^\top}{\sqrt{d}} \right) V_{\text{cross}}, \quad (11)$$

where $Q_{\text{cross}} = \tilde{S} W_{Q_{\text{cross}}}$, $K_{\text{cross}} = \hat{Z}(C_{\text{motion}}) W_{K_{\text{cross}}}$ and $V_{\text{cross}} = \hat{Z}(C_{\text{motion}}) W_{V_{\text{cross}}}$. Finally, the output of MAD can be represented as:

$$\hat{e}_i = \lambda_{\text{spat}} e_i^{\text{spat}} + \lambda_{\text{temp}} e_i^{\text{temp}} + e_i^{\text{cross}}, \quad (12)$$

¹ The architecture of $f_{\text{MAD}}^{\text{proj}}$ is detailed in Appendix C.1

where λ_{spat} and λ_{temp} are weights for e_{spat} and e_{temp} , respectively. The training objectives of MAD are formulated by:

$$\begin{aligned} \mathcal{L}_{\text{MAD}} = & \frac{1}{n} \sum_{i=1}^n |\hat{e}_i - e_i| \\ & - \frac{1}{2n} \sum_{j=1}^n \log \frac{\exp(\text{sim}(\hat{e}_i, e_j)/\tau)}{\sum_{k=1}^n \exp(\text{sim}(\hat{e}_i, e_k)/\tau)} \\ & - \frac{1}{2n} \sum_{j=1}^n \log \frac{\exp(\text{sim}(e_j, \hat{e}_j)/\tau)}{\sum_{k=1}^n \exp(\text{sim}(e_j, \hat{e}_k)/\tau)} \end{aligned} \quad (13)$$

where $\text{sim}(\cdot, \cdot)$ is cosine similarity.

Conditional Video Render (CVR) To fully leverage the information decoded from fMRI signals, we design CVR — a sequential inference framework that reconstructs the video by progressively integrating the fMRI-derived cues at each stage of the generation process. First, the decoded motion latents $\hat{E}(X) = \{\hat{e}_1, \hat{e}_2, \dots, \hat{e}_n\}$ are passed through the pretrained VAE decoder $\mathbf{D}$ to produce a sequence of motion frames $\{I_i^{\text{motion}}\}_{i=1}^n$. Next, the decoded anchor feature $\hat{Z}(C_{\text{anchor}})$ is combined with the first motion frame I_1^{motion} and fed into a text-to-image (T2I) model, yielding the initial reconstructed frame $\hat{v}_1$. Finally, a pretrained text-to-video (T2V) generator is steered jointly by three sources: (i) the holistic semantic guidance $\hat{Z}(C_{\text{holi}})$, (ii) the anchor frame $\hat{v}_1$, and (iii) the motion-frame sequence $\{I_i^{\text{motion}}\}_{i=1}^n$. This combination encourages both temporal smoothness and semantic fidelity, enabling CVR to synthesize a coherent video that closely matches the original visual stimulus inferred from the fMRI signals.

4. Experiments

Datasets and Pre-processing: We evaluate our method on two publicly available fMRI-video datasets: CC2017 [71] and a subset of the HCP 7T dataset [42]. **CC2017** contains fMRI recordings from three subjects (3T MRI) while watching 23 high-definition natural movie clips (8 minutes each, 30 FPS). To improve signal-to-noise ratio (SNR), training clips are repeated twice and test clips ten times; the fMRI responses are averaged accordingly. After aligning

Method	Semantic-Level					Pixel-Level			ST-Level	
	2-way-I $\uparrow$	2-way-V $\uparrow$	50-way-I $\uparrow$	50-way-V $\uparrow$	VIFI-score $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	Hue-pcc $\uparrow$	CLIP $\uparrow$	EPE $\downarrow$
Nishimoto [44](Current Biology'11)	0.742	—	—	—	—	0.119	8.383	0.737	—	—
Wen [71](Cerebral Cortex'17)	0.771	0.070	—	0.166	—	0.130	8.301	0.637	—	—
Kupersmidt [34](arXiv'22)	0.769	0.768	0.179	—	0.591	0.108	<u>11.030</u>	0.583	0.399	6.344
f-CVGAN [66](Cerebral Cortex'22)	0.777	0.721	—	—	0.592	0.108	11.430	0.583	0.399	6.344
Mind-video [10](NeurIPS'23)	0.797	0.848	0.174	0.197	0.593	0.177	8.868	0.768	0.409	6.125
NeuroClips [15](NeurIPS'25)	0.806	0.834	0.203	0.220	<u>0.602</u>	0.390	9.211	0.812	<u>0.513</u>	4.833
Mind-Animator [41](ICLR'25)	0.805	0.830	0.192	0.186	0.608	<u>0.321</u>	9.220	0.786	0.425	5.422
NEURONS [67](ICCV'25)	<u>0.815</u>	<u>0.853</u>	0.213	0.246	0.597	0.285	9.526	<u>0.830</u>	0.482	<u>4.827</u>
Ours	0.838	0.865	0.225	0.264	0.608	<u>0.321</u>	9.448	0.849	0.526	4.788

Table 1. Quantitative comparison of reconstruction results on the CC2017 dataset. All metrics are averaged across all samples and three subjects, with the best results highlighted in bold and the second-best results underlined. 100 sample sets were constructed using the bootstrap method for hypothesis testing. Colors reflect statistical significance compared to our model. $p < 0.0001$ (blue); $p < 0.01$ (pink); $p < 0.05$ (yellow); $p > 0.05$ (green)

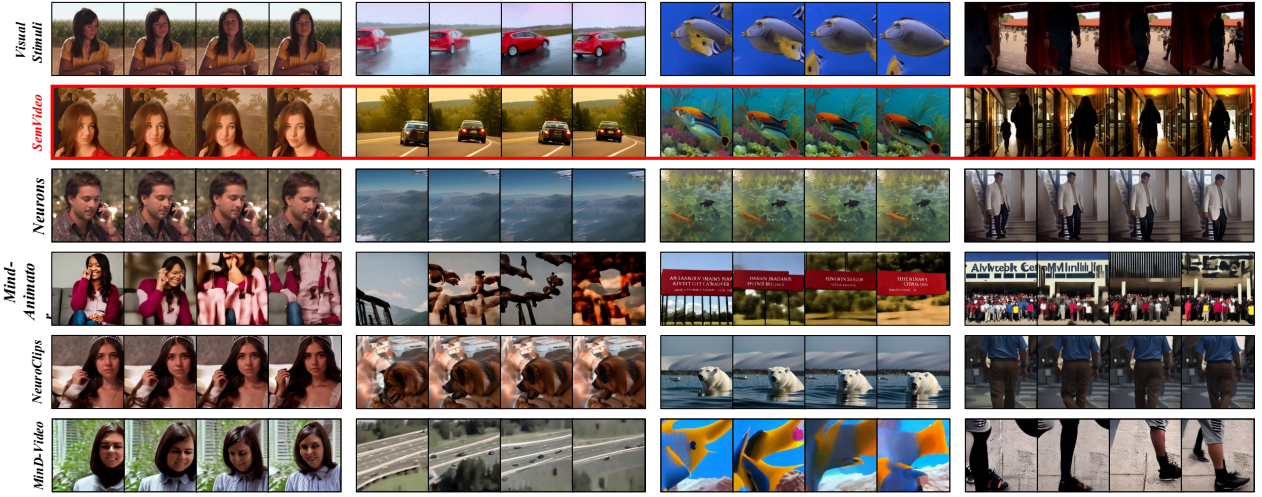


Figure 4. Qualitative comparison of reconstruction results on the CC2017 dataset between previous methods and our proposed *SemVideo*. Reconstructions generated by *SemVideo* are highlighted with red boxes.

Method	Semantic		Pixel		ST-Level	
	2-way-I $\uparrow$	2-way-V $\uparrow$	PSNR $\uparrow$	Hue-pcc $\uparrow$	CLIP $\uparrow$	EPE $\downarrow$
Nishimoto	0.658	—	11.316	0.645	—	—
Wen	0.702	—	10.197	0.727	—	—
Mind-video	0.779	0.769	9.275	0.793	0.499	9.29
Mind-Animator	0.786	<u>0.778</u>	11.233	0.829	<u>0.511</u>	7.08
Ours	0.815	0.796	9.336	0.858	0.523	4.05

Table 2. Quantitative comparison on the HCP dataset.

video frames with the fMRI temporal resolution ($TR = 2s$), we obtain 8,640 training samples and 1,200 test samples. Visually responsive voxels are selected by computing the temporal correlation between repeated presentations, applying Fisher’s z-transform, and conducting a one-sample t-test (Bonferroni corrected, $p < 0.05$). This yields 13,447; 14,828; and 9,114 voxels for the three subjects, respectively. A 4-second BOLD lag is applied to compensate for the

hemodynamic response delay. **HCP 7T**: We use a subset of the HCP 7T dataset, selecting three subjects (IDs: 100610, 102816, 104416) following prior work. The data undergoes HCP’s preprocessing pipeline, including motion/distortion correction, high-pass filtering, ICA denoising, and MNI registration. We extract 5,820 voxels from [14]. We partition clips into training and test sets with a 9:1 split. A 4-second hemodynamic lag is similarly applied.

Evaluation Metrics : We follow previous works that introduce the evaluation metrics at three complementary levels: **Semantic-Level**: (i) *N-way top-1 retrieval* in four settings—2-way/50-way, frame-level [11]/video-level [64]. (ii) *VIFI* [54]: cosine similarity of VIFICLIP features between ground-truth and reconstructions. **Pixel-Level**: SSIM, PSNR, and Hue-PCC [61] quantify low-level fidelity. **Spatiotemporal-Level (ST-Level)**: (i) *CLIP-PCC*:

Method	Semantic-Level					Pixel-Level			ST-Level	
	2-way-I $\uparrow$	2-way-V $\uparrow$	50-way-I $\uparrow$	50-way-V $\uparrow$	VIFI-score $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	Hue-pcc $\uparrow$	CLIP $\uparrow$	EPE $\downarrow$
Ours (full)	0.826	0.860	0.211	0.239	0.590	0.330	9.626	0.841	0.502	4.768
w/o C_{anchor}	0.786	0.808	0.158	0.147	0.534	0.361	9.454	0.835	0.488	4.796
w/o C_{holi}	0.819	0.849	0.205	0.221	0.584	0.299	9.492	0.834	0.490	4.859
w/o C_{motion}	0.818	0.846	0.207	0.216	0.583	0.285	9.353	0.741	0.481	4.930

Table 3. Ablation study evaluating the impact of removing different semantic descriptions— C_{anchor} , C_{holi} , and C_{motion} —from the decoding targets of the Semantic Alignment Decoder (SAD). All results are reported on the CC2017 dataset.

average cosine similarity of CLIP embeddings of adjacent frames; set to 0 when VIFI < 0.6 to avoid spurious inflation. (ii) *EPE* [3]: quantifies motion fidelity by calculating the average Euclidean distance between predicted and ground-truth optical flow vectors at each pixel.

Implementation: The training strategy of the experiment involves pre-training across all subjects, followed by fine-tuning on a single subject. Consistent with prior research [15], we adopt AnimateDiff for T2V task. All models are trained for 100 epochs with AdamW. Loss weights are $\lambda_{SoftCLIP}=0.1$ and $\lambda_{refine}=0.5$. Please refer to Appendix C.1 for more implementation details.

4.1. Main Results

We report the quantitative performance of our method on the CC2017 dataset in Table.1 and qualitative comparisons in Figure.4. *SemVideo* achieves remarkable and comprehensive performance across all evaluation levels.

Superior Semantic Accuracy and Fidelity. Our model outperforms NeuroClips [15] and Neurons [67], achieving a 2-way-V score of 0.865 and a 50-way-V score of 0.264, demonstrating superior semantic consistency and discriminability. On the VIFI-score, which reflects overall semantic-video alignment, *SemVideo* ties with MindAnimator[41] at 0.608, indicating strong semantic fidelity. This quantitative superiority is also reflected visually in Figure.4 where the reconstructions of *SemVideo* are more aligned with the ground truth video stimuli, accurately capturing central objects like “fish” and “cars”.

Competitive Low-Level Reconstruction Quality. For pixel-level reconstruction quality, our model achieves the highest hue-pcc (0.849) and reaches a competitive SSIM of 0.321 and PSNR of 9.448, closely trailing the SoTA method. This high fidelity is also evident visually, as it effectively recovers key low-level features such as lighting and shadow contrasts (e.g., rightmost column in Figure.4).

Enhanced Spatial-Temporal Coherence. Our model achieves the highest CLIP similarity (0.526), and the lowest EPE (4.788), highlighting better preservation of spatial-temporal coherence and alignment with high-level perceptual features. Most notably, this is demonstrated by *SemVideo*’s ability to excel at modeling temporal dy-

namics, successfully reconstructing coherent motion patterns—such as “a person turning their head” (Figure.4, first column)—which remain challenging for previous methods. These results underscore *SemVideo*’s strength in preserving both spatial detail and temporal consistency in fMRI-to-video reconstruction.

Strong Generalization Ability. To assess generalization, we further evaluate on the HCP dataset. As shown in Table.2, *SemVideo* maintains strong performance, demonstrating its robustness across subjects and datasets.

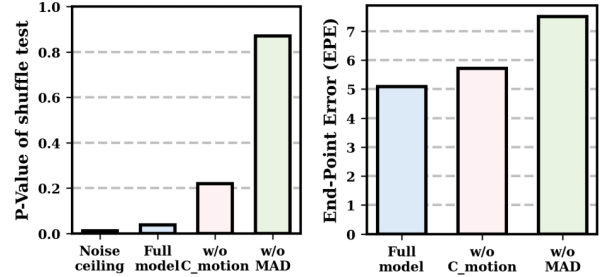


Figure 5. The figure presents two sets of results from subj01 of the CC2017 dataset. On the left, the results of a shuffle test are shown; significance was determined using paired t-tests with Bonferroni correction ($P < 0.05$). On the right, the results of an ablation study are displayed, evaluating the impact of the C_{motion} and MAD guidance components.

4.2. Validating the Source of Motion Improvement.

Following the work of [41], we reconstruct video clips using Tune-a-video [72] which lacks motion priors and conduct shuffle tests to further validate the contributions of both the MAD and the motion-oriented narratives C_{motion} to motion improvement. Specifically, for each 12-frame reconstructed video, we randomly shuffle the order of the frames 100 times and compute the EPE on both the original and shuffled frames. We estimate the P-value using the following formula: $P = \sum_{i=1}^{100} \delta_i / 100$, where $\delta_i = 1$ if the i^{th} shuffle outperforms the reconstruction result in the original order based on the metrics, and $\delta_i = 0$ otherwise. A lower P-value signifies a closer alignment between the sequential order of the reconstructed video and the ground truth. It can be observed that the P-value for our full model is signifi-

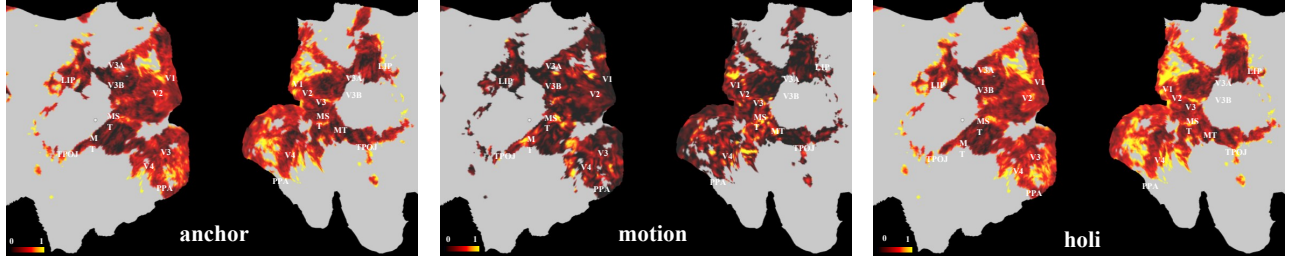


Figure 6. Importance visualization of different ROIs of subject 1 on CC2017, based on the fitted weights from the first layer of the SAD. Weights from each module are averaged and normalized to the $[0, 1]$ range for comparison. Please refer to Appendix D for more subjects’ importance visualization.

cantly low, confirming that its predicted inter-frame order robustly matches the ground truth, but rises substantially when C_{motion} is removed and spikes to 0.84 when the entire MAD is ablated, as illustrated in Figure 5 left. This indicates that we have indeed decoded motion information from MAD and fMRI-derived C_{motion} . This conclusion is reinforced by a comparison of the EPE score, as shown in Figure 5 right. The EPE increase when removing C_{motion} and the spike when ablating the MAD confirm the crucial role of motion semantic guidance and that the MAD , not T2V priors, is the source of decoded motion improvement.

4.3. Ablation Study

In this section, we conduct a detailed ablation study to evaluate the individual contributions of the C_{anchor} , C_{holi} , and C_{motion} components to our *SemVideo*. The quantitative results, presented in Table 3, unequivocally demonstrate that the removal of any single component leads to a performance degradation across nearly all metrics, affirming their indispensability. Specifically, the exclusion of C_{motion} causes the most precipitous decline in pixel-level and ST-level metrics. The absence of C_{anchor} , in contrast, significantly impairs semantic-level fidelity. The C_{holi} proves integral for overall performance, as its removal adversely affects all metrics, underscoring its global guidance under video reconstruction. Crucially, and in contrast to prior works employing conventional latent decoders, the removal of C_{motion} also results in a notable drop in semantic metrics. This key finding substantiates the efficacy of our multi-level semantic guidance mechanism. It demonstrates that our motion latent space is not only responsible for ensuring pixel-level consistency and temporal smoothness but is also endowed with semantic understanding, thereby guaranteeing the coherence of the synthesized actions.

4.4. Neuroscience Interpretability

To investigate the neural interpretability of our model, we generated voxel-wise importance maps over the cerebral cortex. The results, presented in Figure 6, shed light on the neural substrates underlying each core component. The an-

chor component (Figure 6, left) predominantly draws from a hierarchy of visual regions, with a notable emphasis on higher-level visual cortices, confirming its role in encoding abstract scene structure. The motion component (Figure 6, middle) aligns well with established motion-processing regions, including MT, MST, and TPOJ [5]. Interestingly, it also shows strong activations in V1, supporting the existence of parallel, non-hierarchical pathways—where V1 may directly convey motion signals to MT [43]—in contrast to the traditional view of a strictly feedforward dorsal stream [78]. Lastly, the holistic component (Figure 6, right) exhibits a distributed pattern of importance across both visual and motion-related areas. This balanced integration highlights its function as a global information aggregator, responsible for synthesizing perceptual and temporal features to support coherent reconstruction. This hierarchical semantic mapping enables *SemVideo* to generate more perceptually faithful reconstructions and similar activation distributions across subjects demonstrate stable feature capture for both static semantics and dynamic behaviors.

5. Conclusion

Reconstructing dynamic visual stimuli from brain activity remains a challenge in cognitive neuroscience. In this work, we propose *SemVideo*, an innovative fMRI-to-video reconstruction framework featuring a novel semantic alignment decoder, a motion adaptation decoder and a conditional video render. By leveraging multi-level semantic cues, *SemVideo* effectively addresses the challenges of both semantic sparsity and temporal coherence in video reconstruction. Our experimental evaluation demonstrates the superior performance of *SemVideo* on two public datasets, surpassing existing methods across multiple metrics, including semantic, pixel, and spatiotemporal levels. Furthermore, our use of ROI-wise visualization techniques provides valuable neuroscience-driven insights into the effectiveness of the proposed method. We believe *SemVideo* will set a new SOTA in fMRI-to-video decoding and lay the foundation for future brain-based video reconstruction.

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